Algebra 1
Unit 2 Assessment

Name______________________
Date__________________
Period__________

Generate an equivalent algebraic expression where each variable only appears once.

1. $(5h^{-6}m^3)(6h^5m^7)^2$

work

Equivalent Algebraic Expression	$180h^4m^{17}$

2. $(-7xy^{-4})(9x^5z^3)(3x^4y^2z^{-1})$

work

Equivalent Algebraic Expression	$-189x^{10}y^{-2}z^2$ or $-\frac{189x^{10}z^2}{y^2}$

3. $(-8h^3k^{-5})^2 \cdot (10h^0k^6)$

work

Equivalent Algebraic Expression	$640h^6k^{-4}$ or $\frac{640h^6}{k^4}$

4. $\frac{(2c^3)^5(4c^4)}{(8c^6)^2}$

work		Equivalent Algebraic Expression	$2c^7$

5. Use the laws of exponents to complete the equivalent expression for 5.6^{2x}.

$$\left(\boxed{31.36}\right)^x$$

6. Use the laws of exponents to complete the equivalent expression for 3^{5x-3}.

$$\frac{\left(\boxed{243}\right)^x}{\boxed{27}}$$

7. Find the sum.

$$(x+3)(2x^5)+7x^5$$

work

Sum	$2x^6+13x^5$

8. Find the product.

$$(k+8)(k-11)$$

work

Product	$k^2-3k-88$

9. Find the sum.

$$-2x(x+9)+(x+8)(7-3x)$$

work

Sum	$-5x^2-35x+56$

10. Find the difference.

$$\frac{2}{3}(7y^3-9y)-\left(\frac{11}{3}y^3-\frac{2}{3}y^2+\frac{17}{3}\right)$$

work

Difference	$y^3+\frac{2}{3}y^2-6y-\frac{17}{3}$

11. Find the product.

$$(2x+9)(5x-8)$$

work

Product	$10x^2+29x-72$

12. Find the product.

$$(x+7)^2$$

work

Product	$x^2+14x+49$

13. Find the product.

$$\frac{4}{5}b^6(11b - 15)$$

work

Product	$\frac{44}{5}b^7 - 12b^6$

14. Find the quotient where $h \neq 0$.

$$(24h^9 + 32h^{15}) \div (-8h^3)$$

work

Quotient	$-3h^6 - 4h^{12}$

15. Find the product.

$$(x-4)(x-7)$$

work

Product	$x^2 - 11x + 28$

16. Find the quotient where $w \neq 0$ and $y \neq 0$.

$$\frac{16w^5y^{11} + 28w^2y^5 - 36w^3y^3}{4wy^3}$$

work

Quotient	$4w^4y^8 + 7wy^2 - 9w^2$